Database Management System (DBMS)

Previous Year Question Paper – 2022
Time: 3 Hours Maximum Marks: 100

Note: This is a sample/dummy placeholder question paper generated for demonstration purposes only. It does not represent an official examination paper. Some questions repeat across years with different wording, as is common in real previous-year question compilations.

Section A – Short Answer Questions (Answer any 5)

Q1. Describe the ACID properties of a transaction. *(5 marks each)*

Q2. What is a primary key? How is it different from a candidate key? *(5 marks each)*

Q3. Define a foreign key and explain referential integrity constraint. *(5 marks each)*

Q4. What is a transaction? Explain the transaction states with a diagram. *(5 marks each)*

Q5. Differentiate between BCNF and 3NF. *(5 marks each)*

Section B – Medium Answer Questions (Answer any 4)

Q1. What is indexing in DBMS? Explain B-Tree and B+ Tree indexing. *(10 marks each)*

Q2. Discuss deadlock in DBMS. Explain deadlock detection and prevention techniques. *(10 marks each)*

Q3. Explain the concept of views in SQL with an example. *(10 marks each)*

Q4. Explain the different types of relationships (one-to-one, one-to-many, many-to-many) with ER diagram examples. *(10 marks each)*

Section C – Long Answer Questions (Answer any 3)

Q1. Explain the E-R to relational mapping process with an example. *(15 marks each)*

Q2. Discuss distributed databases. Explain their advantages and challenges. *(15 marks each)*

Q3. Design a relational schema for a Hospital Management System. Identify entities, attributes, and relationships. Convert the ER diagram into relational tables. *(15 marks each)*

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